

SARNA, Sweden

WMO: 023160

Lat: 61.691N

Lon: 13.187E

Elev: 425

StdP: 96.32

Time Zone: 1.00 (EUC)

Period: 86-99

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-31.1	-27.7	-33.5	0.2	-31.0	-30.1	0.2	-27.7	12.3	0.6	10.4	-5.7	0.5	320	0.696

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.8	24.2	15.3	22.2	14.0	20.3	13.1	16.2	22.0	15.1	20.4	14.0	18.7	2.6	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
13.9	10.5	17.4	12.7	9.6	16.7	11.5	8.9	15.8	46.7	21.9	43.4	20.6	40.5	18.8	19.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.0	7.3	6.8	-34.7	26.8	4.2	1.7	-37.7	28.0	-40.2	29.1	-42.6	30.0	-45.6	31.3		
(5)			-34.8	17.5	4.2	1.4	-37.8	18.5	-40.2	19.4	-42.6	20.2	-45.6	21.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	1.0	-10.3	-8.9	-4.6	-0.1	6.0	11.0	13.2	11.3	6.8	1.7	-5.0	-10.1	(6)
(7)		DBStd	9.83	8.23	7.41	5.38	4.03	3.72	3.13	2.99	3.34	3.14	4.14	5.78	7.57	(7)
(8)		HDD10.0	3537	629	529	454	302	133	23	6	24	105	258	451	623	(8)
(9)		HDD18.3	6340	887	762	712	552	381	221	163	218	346	516	701	881	(9)
(10)		CDD10.0	242	0	0	0	0	11	53	104	65	9	0	0	0	(10)
(11)		CDD18.3	3	0	0	0	0	0	1	2	1	0	0	0	0	(11)
(12)		CDH23.3	88	0	0	0	0	3	27	42	16	0	0	0	0	(12)
(13)		CDH26.7	4	0	0	0	0	0	3	2	0	0	0	0	0	(13)
(14)	Wind	WSAvg	1.9	1.8	2.0	2.1	1.9	2.1	2.1	1.9	1.9	1.8	1.5	1.7	(14)	
(15)	Precipitation	PrecAvg	666	41	29	28	40	56	79	84	96	63	61	52	38	(15)
(16)		PrecMax	878	88	54	58	104	115	164	140	161	112	135	139	65	(16)
(17)		PrecMin	545	7	6	4	5	7	5	7	48	22	16	19	4	(17)
(18)		PrecStd	79	21	12	15	28	30	32	30	34	26	32	28	15	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.1	6.6	8.5	15.5	23.8	26.4	26.7	25.1	19.2	12.5	6.9	4.2	(19)
(20)			MCWB	2.5	3.3	4.8	8.6	13.4	17.4	16.8	15.6	13.5	9.1	4.8	2.5	(20)
(21)		2%	DB	3.2	4.7	6.0	11.7	21.3	22.9	24.4	23.1	16.4	10.5	5.2	3.0	(21)
(22)			MCWB	1.1	2.3	2.3	6.2	12.2	14.6	15.6	15.0	10.9	7.7	3.7	1.5	(22)
(23)		5%	DB	1.5	2.9	4.1	8.8	18.2	20.6	22.1	20.5	14.3	9.3	3.5	1.7	(23)
(24)			MCWB	0.0	1.0	1.2	4.2	10.5	12.9	14.3	13.8	10.0	6.9	2.4	0.4	(24)
(25)		10%	DB	0.4	1.2	2.7	6.7	15.0	18.3	20.2	17.9	12.6	7.8	1.8	0.4	(25)
(26)			MCWB	-0.8	-0.4	0.5	3.2	8.7	11.8	13.6	12.8	9.1	5.8	0.9	-0.6	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.8	3.9	5.4	9.1	14.3	17.6	17.7	16.9	14.0	10.0	5.1	3.0	(27)
(28)			MCD B	4.6	5.8	7.9	14.8	21.9	25.6	24.0	22.5	17.3	11.5	6.2	3.8	(28)
(29)		2%	WB	1.4	2.5	2.9	6.5	12.6	15.5	16.3	15.6	12.1	8.4	3.8	1.8	(29)
(30)			MCD B	2.9	4.4	4.8	11.0	20.1	21.2	22.3	21.0	14.7	9.8	5.0	2.8	(30)
(31)		5%	WB	0.3	1.1	1.7	4.8	11.0	13.8	15.2	14.5	10.8	7.1	2.4	0.5	(31)
(32)			MCD B	1.3	2.8	3.5	7.8	17.4	19.1	20.4	19.3	13.5	8.9	3.4	1.5	(32)
(33)		10%	WB	-0.7	-0.2	0.6	3.6	9.2	12.4	14.2	13.3	9.4	6.0	1.0	-0.5	(33)
(34)			MCD B	0.3	1.0	2.3	6.3	14.6	17.2	18.9	17.2	12.1	7.5	1.7	0.4	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	9.0	10.7	10.6	10.0	12.1	10.6	10.8	10.3	9.7	7.3	6.7	8.9	(35)
(36)			MCDBR	8.9	8.1	10.6	14.1	19.4	16.2	15.3	14.7	12.6	9.5	7.1	9.8	(36)
(37)			MCWBR	8.0	6.7	7.6	8.9	11.0	8.8	7.5	7.6	7.7	7.0	6.1	8.6	(37)
(38)		5% WB	MCD BR	8.6	8.1	9.1	12.1	18.8	14.2	12.7	12.3	9.4	7.8	6.2	9.2	(38)
(39)			MCWBR	8.0	7.0	6.8	7.9	10.7	8.1	6.7	6.6	5.9	6.3	5.5	8.3	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.212	0.263	0.291	0.327	0.337	0.332	0.348	0.339	0.308	0.292	0.230	0.341	(40)	
(41)		taud	2.124	2.266	2.272	2.339	2.458	2.499	2.464	2.481	2.551	2.491	2.212	2.824	(41)	
(42)		Ebn at Noon	524	725	827	865	886	896	869	848	818	697	494	372	(42)	
(43)		Edh at Noon	34	61	87	100	97	95	96	87	68	50	31	21	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.27	0.93	2.48	4.12	4.70	5.17	4.70	3.65	2.37	1.06	0.33	0.14	(44)	
(45)		RadStd	0.03	0.17	0.24	0.39	0.43	0.44	0.61	0.33	0.28	0.18	0.08	0.02	(45)	

Historical Trends

		Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (47 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page